



(Somaiya Vidyavihar University)

Batch: A2 Roll No.: 16010324044

Experiment No :- 7

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of the Staff In-charge with date

TITLE: Design of Finite Impulse Response (FIR) (Windowing Method)

AIM: To design FIR filter for the given specifications and observe the magnitude and phase response

OUTCOME: Design and realize Digital Filters, Design Finite Impulse Response filters

PROCEDURE:

1. Design an FIR filter for the given specifications and observe the magnitude and phase response of the designed filter.
2. Change the filter specifications and observe the effect on the order of the filter.

Design:

Design a 11-tap linear phase FIR low pass filter with a cut-off frequency of 100 Hz and a sampling rate of 400 Hz using the Hamming/Hanning window function. Determine the transfer function and difference equation of the designed FIR system. Compute and plot the magnitude and phase response for $\omega=0$, $\omega=\pi/4$, $\omega=\pi/2$ and $\omega=\pi$ radians.

Calculations:

$$\omega_c = 2\pi f_c T_s =$$

$$\text{For lowpass filter } h(n) = \begin{cases} \frac{\omega_c}{\pi} & n = 0 \\ \frac{\sin(\omega_c n)}{n\pi} & n \neq 0 \end{cases} \quad -M \leq n \leq M$$

$$\begin{aligned} h(0) = & & h(1) = h(-1) = & & h(2) = h(-2) = \\ h(3) = h(-3) = & & h(4) = h(-4) = & & h(5) = h(-5) = \end{aligned}$$

$$w_{ham}(n) = 0.54 + 0.46 \cos\left(\frac{n2\pi}{M}\right)$$

$$\begin{aligned} w_{ham}(0) = & & w_{ham}(1) = & & w_{ham}(2) = \\ w_{ham}(3) = & & w_{ham}(4) = & & w_{ham}(5) = \\ w_{ham}(-1) = & & w_{ham}(-2) = & & \\ w_{ham}(-3) = & & w_{ham}(-4) = & & w_{ham}(-5) = \end{aligned}$$

$$\begin{aligned} h_w(n) &= h(n) \times w_{ham}(n) \\ h_w(0) = & & h_w(1) = & & h_w(2) = & & h_w(3) = \\ h_w(4) = & & h_w(5) = & & h_w(-1) = & & \\ h_w(-2) = & & h_w(-3) = & & h_w(-4) = & & h_w(-5) = \end{aligned}$$

$$H(z) =$$

$$H(e^{j\omega}) =$$

$$|H(e^{j\omega})| =$$

$$\angle H(e^{j\omega}) =$$

Quantized coefficients to fixed point word length 4

Code:

```

clc;
clear all;
close all;
%% FIR Filter Specifications
Fs = 1000;    % Sampling frequency in Hz
Fp = 120;     % Passband edge frequency in Hz
Fst = 150;    % Stopband edge frequency in Hz
Ast = 80;     % Desired stopband attenuation in dB
% Transition width
Df = Fst - Fp;
% Cut-off frequency
Fc = (Fp + Fst)/2;
% Normalized cut-off frequency
Wn = Fc/(Fs/2);
%% Filter Orders for Different Windows
N_rect = ceil(0.9*Fs/Df); % Rectangular Window
N_hann = ceil(3.1*Fs/Df); % Hanning Window
N_hamm = ceil(3.3*Fs/Df); % Hamming Window
% Kaiser Window
N_kais = ceil((Ast-8)/(2.285*2*pi*Df/Fs));
% Make all orders even
N_rect = N_rect + mod(N_rect,2);
N_hann = N_hann + mod(N_hann,2);
N_hamm = N_hamm + mod(N_hamm,2);
N_kais = N_kais + mod(N_kais,2);
%% Kaiser beta calculation
if Ast > 50
    beta = 0.1102*(Ast-8.7);
elseif Ast >= 21
    beta = 0.5842*(Ast-21)^0.4 + 0.07886*(Ast-21);
else
    beta = 0;
end
%% FIR Filter Design
% Rectangular window
b_rect = fir1(N_rect, Wn, 'low', rectwin(N_rect+1));
% Hanning window
b_hann = fir1(N_hann, Wn, 'low', hann(N_hann+1));
% Hamming window
b_hamm = fir1(N_hamm, Wn, 'low', hamming(N_hamm+1));
% Kaiser window
b_kais = fir1(N_kais, Wn, 'low', kaiser(N_kais+1,beta));
%% Display Filter Orders
fprintf('===== \n');
fprintf('    FIR FILTER ORDERS \n');
fprintf('===== \n');
fprintf('Rectangular Window : %d \n', N_rect);
fprintf('Hanning Window    : %d \n', N_hann);
fprintf('Hamming Window     : %d \n', N_hamm);
fprintf('Kaiser Window      : %d \n', N_kais);
fprintf('===== \n');

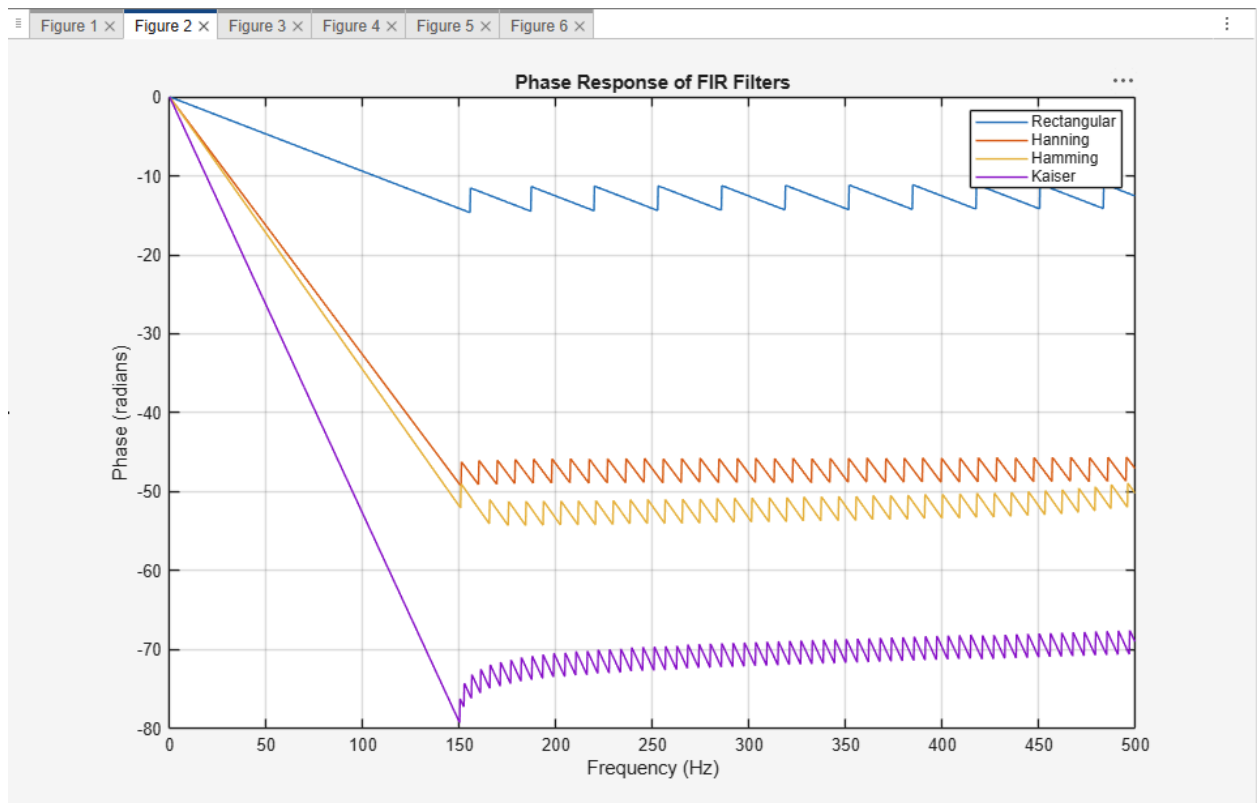
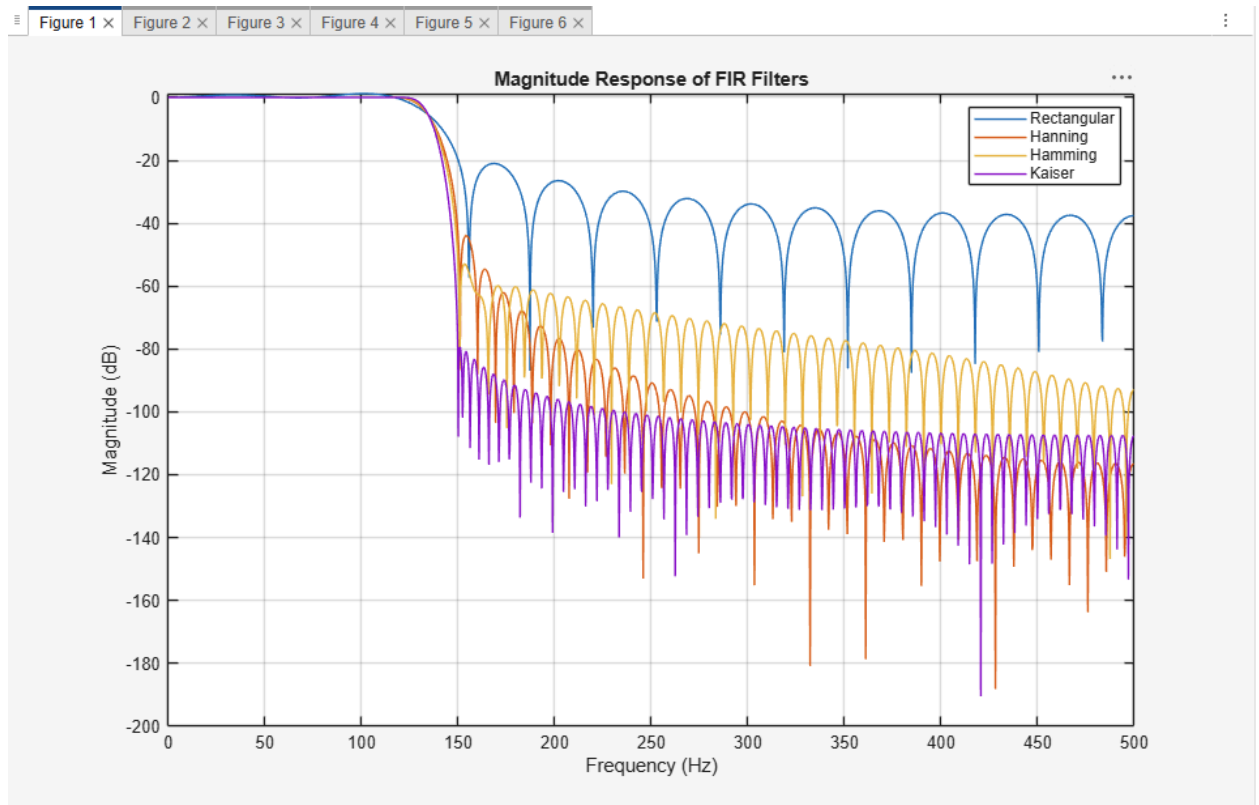
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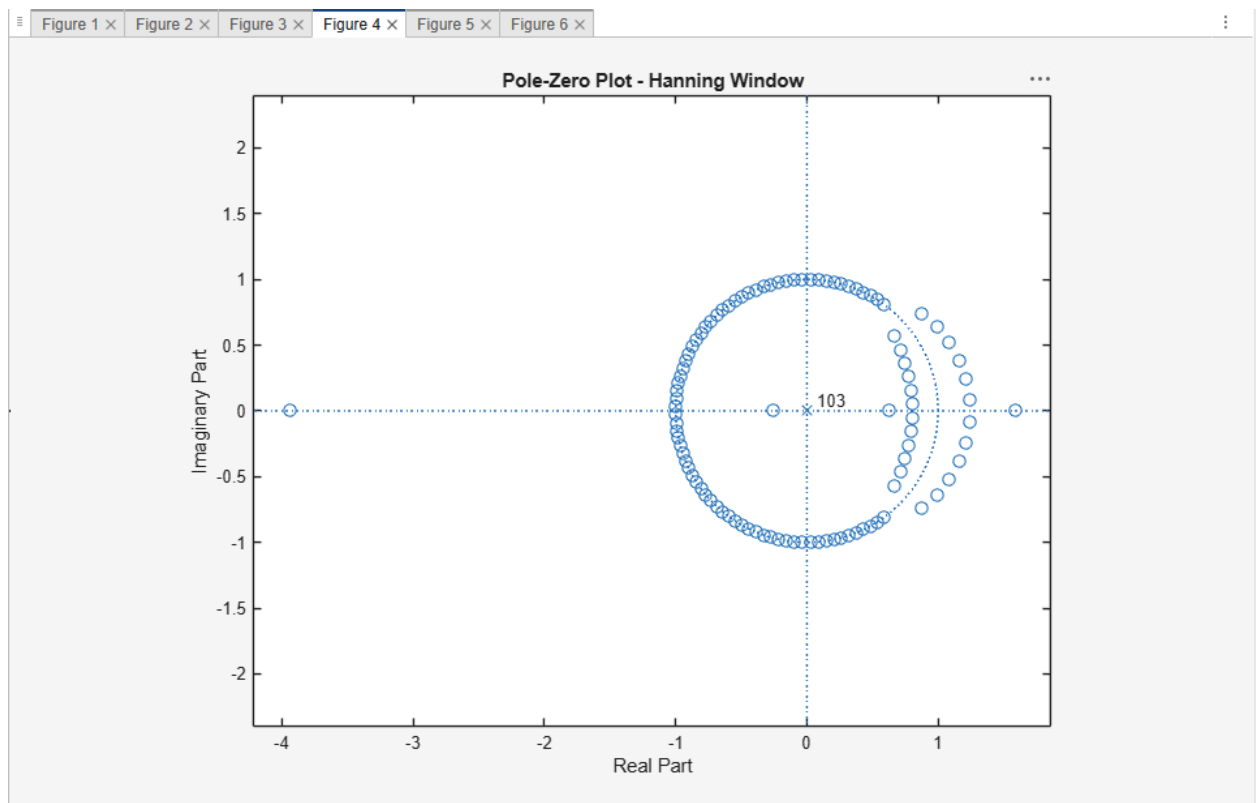
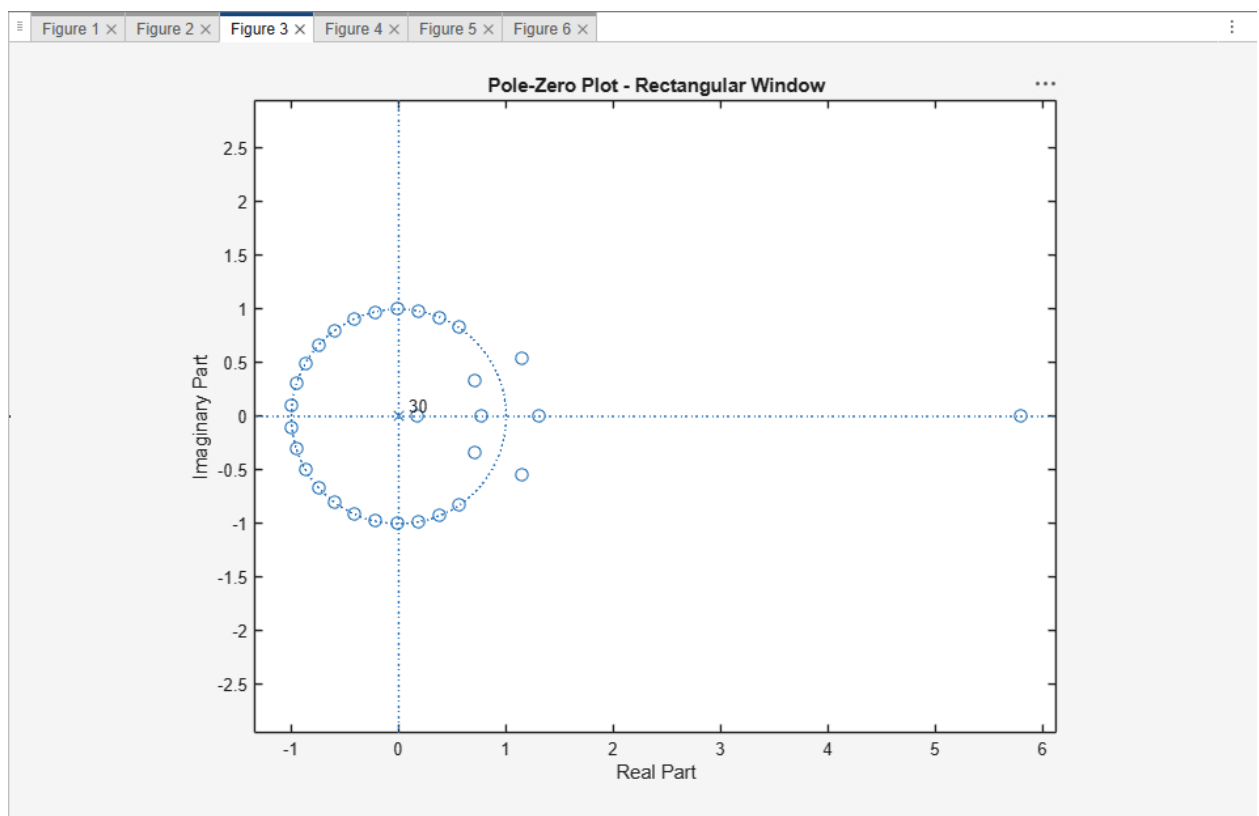
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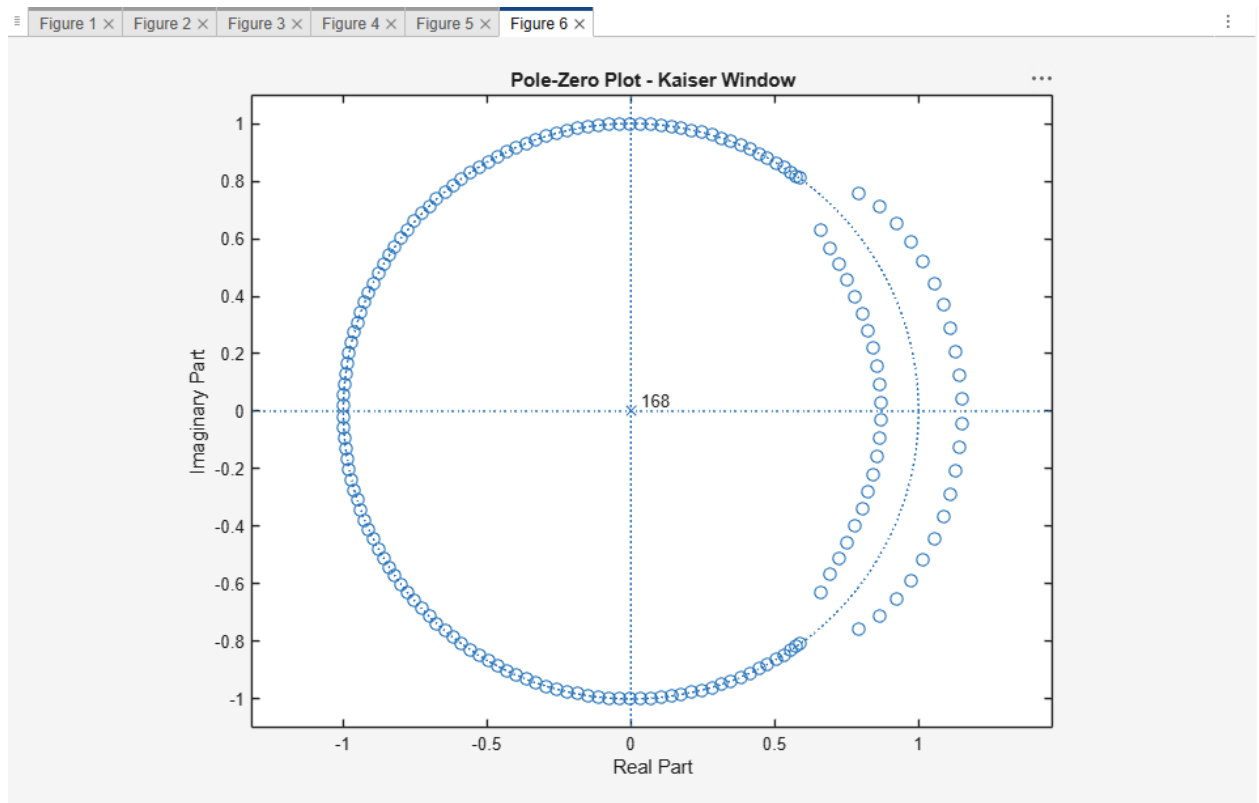
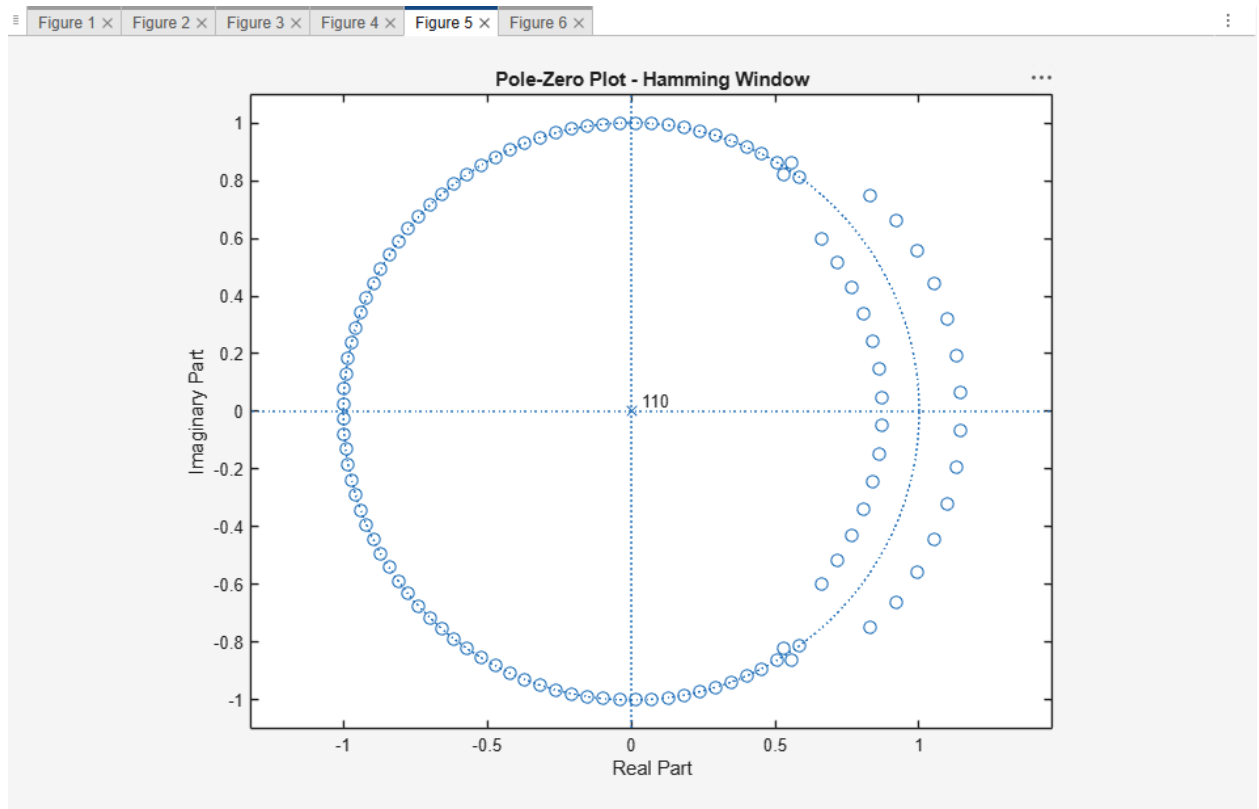
%% Frequency Response
[H_rect,f] = freqz(b_rect,1,2048,Fs);
[H_hann,f] = freqz(b_hann,1,2048,Fs);
[H_hamm,f] = freqz(b_hamm,1,2048,Fs);
[H_kais,f] = freqz(b_kais,1,2048,Fs);
%% Magnitude Response
figure;
plot(f,20*log10(abs(H_rect)));
hold on;
plot(f,20*log10(abs(H_hann)));
plot(f,20*log10(abs(H_hamm)));
plot(f,20*log10(abs(H_kais)));
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Magnitude Response of FIR Filters');
legend('Rectangular','Hanning','Hamming','Kaiser');
xlim([0 Fs/2]);
%% Phase Response
figure;
plot(f,unwrap(angle(H_rect)));
hold on;
plot(f,unwrap(angle(H_hann)));
plot(f,unwrap(angle(H_hamm)));
plot(f,unwrap(angle(H_kais)));
grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (radians)');
title('Phase Response of FIR Filters');
legend('Rectangular','Hanning','Hamming','Kaiser');
xlim([0 Fs/2]);
%% Pole-Zero Plot
figure;
zplane(b_rect,1);
title('Pole-Zero Plot - Rectangular Window');
figure;
zplane(b_hann,1);
title('Pole-Zero Plot - Hanning Window');
figure;
zplane(b_hamm,1);
title('Pole-Zero Plot - Hamming Window');
figure;
zplane(b_kais,1);
title('Pole-Zero Plot - Kaiser Window');

```

Output:







Application:

ω radians	f Hz	CALCULATED			OBSERVED	
		$ H(e^{j\omega}) $	$ H(e^{j\omega}) _{dB}$	$\angle H(e^{j\omega})$	$ H(e^{j\omega}) _{dB}$	$\angle H(e^{j\omega})$
0						
$\frac{\pi}{4}$						
$\frac{\pi}{2}$						
$\frac{3\pi}{4}$						
π						

Realization of filter:

Observation:

For an FIR filter, the denominator is 1, so the poles are located at the origin.

For the changed specifications:

Window	Filter Order	Poles at Origin
Rectangular	30	30
Hanning	104	104
Hamming	110	110
Kaiser	168	168

The zeros are distributed around the unit circle and determine the attenuation characteristics of the filter.

Rectangular Window:

Zeros are distributed around the unit circle and produce relatively high sidelobes.

Hanning Window:

Zeros are more densely distributed around the unit circle, giving improved sidelobe suppression.

Hamming Window:

The zero distribution provides better stopband attenuation and reduced sidelobes.

Kaiser Window:

The larger number of zeros provides stronger stopband suppression and allows the required attenuation to be achieved.

The completed reference contains the corresponding four pole-zero plots.

CONCLUSION

The FIR low-pass filter was successfully designed using the windowing method. The magnitude and

phase responses were observed, confirming the linear-phase characteristics of the FIR filter. It was also observed that increasing attenuation requirements and narrowing the transition band increases the filter order.

Signature of faculty in-charge